

GILT EDGE NORTH AGCM, AB, Canada

WMO: 717910

Lat: 53.0710N

Lon: 110.6211W

Elev: 647

StdP: 93.79

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.5	-28.4	-35.0	0.1	-31.4	-31.7	0.2	-28.3	13.4	-1.7	11.7	-3.3	2.5	260	0.698

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	28.6	17.7	26.8	16.7	25.1	16.3	19.5	25.9	18.3	24.5	17.2	23.0	4.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.3	22.6	16.0	12.3	21.3	14.9	11.5	20.1	58.6	25.8	54.6	24.4	51.1	23.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.0	8.8	DB	-35.7	31.8	2.2	2.0	-37.2	33.2	-38.5	34.4	-39.7	35.6	-41.3	37.0	(4)
(5)			WB	-35.8	21.6	2.2	1.4	-37.3	22.7	-38.6	23.5	-39.8	24.3	-41.4	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-11.6	-12.2	-5.1	3.1	11.0	15.0	17.5	16.2	11.5	4.3	-4.9	-12.6	(6)
(7)		DBStd	12.62	8.60	7.99	7.61	5.34	4.45	3.19	2.87	3.20	4.61	4.98	7.10	8.02	(7)
(8)		HDD10.0	3389	669	621	468	212	42	4	0	1	36	187	448	701	(8)
(9)		HDD18.3	5731	928	854	727	456	228	106	51	80	207	437	698	959	(9)
(10)		CDD10.0	752	0	0	0	6	73	156	232	193	82	9	0	0	(10)
(11)		CDD18.3	51	0	0	0	0	2	7	25	14	3	0	0	0	(11)
(12)		CDH23.3	796	0	0	0	3	68	103	304	217	99	3	0	0	(12)
(13)		CDH26.7	173	0	0	0	0	8	16	75	46	28	0	0	0	(13)
(14)	Wind	WSAvg	4.0	4.1	3.9	4.2	4.7	4.3	3.9	3.6	3.4	3.6	4.1	3.9	3.8	(14)
(15)	Precipitation	PrecAvg	416	18	13	16	29	43	73	79	58	35	18	16	19	(15)
(16)		PrecMax	605	42	27	44	91	101	132	166	148	97	38	54	33	(16)
(17)		PrecMin	223	5	4	4	2	6	15	23	9	2	2	2	7	(17)
(18)		PrecStd	88	10	6	9	21	26	30	38	25	24	10	12	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	5.6	12.7	22.1	27.6	28.7	31.7	30.2	30.2	22.5	13.2	5.2	(19)
(20)			MCWB	2.7	2.7	6.3	10.2	13.7	17.3	19.5	17.8	16.0	12.7	6.3	1.9	(20)
(21)		2%	DB	4.2	3.2	9.3	18.4	25.5	26.1	28.7	27.7	26.2	17.3	8.6	3.2	(21)
(22)			MCWB	1.6	1.0	4.7	8.4	13.0	15.8	19.3	17.4	14.7	10.1	4.4	0.7	(22)
(23)		5%	DB	2.6	1.2	6.2	15.0	22.6	24.1	26.6	25.8	23.1	14.5	5.6	0.7	(23)
(24)			MCWB	0.5	-0.3	3.0	6.9	11.8	15.1	18.2	17.2	13.8	8.3	2.4	-1.3	(24)
(25)		10%	DB	0.4	-0.3	3.9	12.1	20.3	22.2	24.7	23.8	20.2	12.0	3.4	-1.7	(25)
(26)			MCWB	-1.1	-1.7	1.4	5.4	10.7	14.3	17.4	16.3	12.5	7.0	1.1	-3.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	3.0	7.3	10.6	14.9	18.8	21.9	20.5	17.2	13.2	7.4	2.4	(27)
(28)			MCDWB	5.4	5.4	12.3	21.6	25.1	25.3	28.7	26.5	26.7	21.3	11.7	4.9	(28)
(29)		2%	WB	1.8	1.1	4.8	8.7	13.6	17.1	20.1	19.0	15.6	11.0	4.9	0.8	(29)
(30)			MCDWB	4.0	2.6	8.7	17.1	22.8	23.5	26.6	25.0	24.5	15.7	8.3	3.2	(30)
(31)		5%	WB	0.5	-0.2	3.1	7.2	12.5	16.1	19.0	17.8	14.3	9.1	2.8	-1.2	(31)
(32)			MCDWB	2.5	1.2	6.3	14.2	21.0	21.8	25.1	23.9	22.0	13.8	5.1	0.5	(32)
(33)		10%	WB	-1.1	-1.8	1.6	5.8	11.4	15.1	17.9	16.8	13.0	7.4	1.3	-3.2	(33)
(34)			MCDWB	0.4	-0.3	3.7	11.2	19.2	20.9	23.7	22.5	19.6	11.8	3.2	-1.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	9.3	9.2	10.9	12.9	11.4	11.8	12.1	12.6	10.4	7.9	7.8	(35)
(36)			MCDBR	9.5	10.4	11.4	16.2	16.6	14.4	15.0	16.1	18.2	15.4	10.7	10.4	(36)
(37)			MCWBR	7.7	8.7	7.7	8.2	7.4	6.4	6.8	7.4	8.2	8.6	7.1	8.7	(37)
(38)		5% WB	MCDWR	9.5	9.7	11.1	15.2	15.2	12.6	13.5	13.8	16.7	14.4	10.0	10.1	(38)
(39)			MCWBR	7.9	8.3	7.6	8.2	7.1	6.3	6.8	6.9	7.9	8.3	7.1	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.246	0.294	0.284	0.330	0.352	0.371	0.364	0.376	0.314	0.278	0.251	0.225	(40)
(41)		taud		2.336	2.208	2.404	2.363	2.351	2.328	2.375	2.319	2.470	2.543	2.452	2.361	(41)
(42)		Ebn at Noon		765	805	915	904	897	879	879	844	870	843	762	734	(42)
(43)		Edh at Noon		56	89	90	108	115	120	112	112	85	63	50	45	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.03	2.09	3.57	4.71	5.81	5.96	6.22	5.06	3.57	2.09	1.03	0.77	(44)
(45)		RadStd		0.10	0.15	0.28	0.43	0.40	0.42	0.33	0.39	0.38	0.29	0.11	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page